

Question 1: How reliable is the communication between the devices? Do you experience any packet loss or delays? If so, under what conditions?

The communication between the client and the server was highly reliable during the experiment. As shown in the reliability plot, 100% of the CoAP requests sent by the client were received by the server and 100% of the server responses were successfully acknowledged by the client.

No packet loss, failed CoAP transactions or communication interruptions were observed during the measurement period. Every request resulted in a corresponding CoAP ACK response.

The measurements were performed under normal laboratory conditions with a short distance between the devices and without intentional interference.

Result:

- Packet loss: 0%
- Successful request delivery: 100%
- Successful response delivery: 100%

Question 2: What is the average round-trip time (RTT) between a CoAP request and the corresponding response?

The RTT measurements show a very stable communication latency. Most requests resulted in an RTT of approximately 14 ms, while only a few measurements reached 15 ms.

The RTT-over-time plot demonstrates that the latency remained nearly constant throughout the entire experiment and no significant timing spikes were observed.

Based on the collected measurements:

- Minimum RTT: 13 ms
- Maximum RTT: 15 ms
- Average RTT: approximately 14 ms
- RTT variation: +- 1 ms

This indicates a deterministic and predictable communication behavior.

Question 3: Does the triggering of the action endpoint work reliably? Does it feel responsive?

The action endpoint operated reliably during all performed tests.

Whenever the button was pressed on the client device, a POST request was sent to the server and the server successfully toggled the internal action state. The corresponding response was received by the client without any failed transactions.

No noticeable delay was perceived between pressing the button and receiving the response. Since the measured RTT is approximately 14 ms, the network-induced delay is effectively imperceptible to the user.

Question 4: How does IEEE 802.15.4 + CoAP compare conceptually to BLE communication regarding complexity, reliability and responsiveness?

A quantitative comparison was not performed because the previous BLE implementation was only evaluated in Renode, whereas the IEEE 802.15.4 + CoAP implementation was evaluated on real hardware.

Conceptually, IEEE 802.15.4 combined with IPv6, UDP and CoAP provides a more Internet-oriented architecture. Devices can communicate using standard IP-based protocols, which simplifies integration into larger IoT networks.

BLE communication is generally simpler to set up for point-to-point communication and typically consumes less protocol overhead. However, it does not natively provide IP connectivity in the same way as a 6LoWPAN-based IEEE 802.15.4 network.

Based on the measurements performed in this lab, the IEEE 802.15.4 + CoAP solution demonstrated reliable communication and low latency on real hardware.